

INMO(5th) – 1990

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Given the equation $x^4 + px^3 + qx^2 + rx + s = 0$ has four real, positive roots, prove that:
 - (a) $pr - 16s \geq 0$.
 - (b) $q^2 - 36s \geq 0$

With equality in each case holding if and only if the four roots are equal.

2. Determine all non-negative integral pairs (x, y) for which $(xy - 7)^2 = x^2 + y^2$.
3. Let f be a function defined on the set of non-negative integers and taking values in the same set. Given that:
 - (a) $x - f(x) = 19[x/19] - 90[f(x)/90]$ for all non-negative integers x .
 - (b) $1900 < f(1990) < 2000$.

Find all possible values that $f(1990)$ can take. Here $[z]$ means greatest integer function.

4. Consider the collection of all three-element subsets drawn from the set $\{1, 2, 3, 4, \dots, 299, 300\}$. Determine the number of those subsets for which the sum of the elements is a multiple of 3.
5. Let a, b, c denote the sides of a triangle, show that the quantity $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$ must lie between the limits $3/2$ and 2 . Can equality hold at either limits?
6. Triangle ABC is scalene with angle A having a measure greater than 90 degrees. Determine the set of points D that lie on the extended line BC , for which $|AD| = \sqrt{|BD| \cdot |CD|}$ where $|BD|$ refers to the (positive) distance between B and D .
7. Let ABC be an arbitrary acute angled triangle. For any point P lying within the triangle, let D, E, F denote the feet of the perpendiculars from P to sides AB, BC and CA respectively. Determine the set of all possible positions of the point P for which the triangle DEF is isosceles. For which position of P will the triangle DEF become equilateral?

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